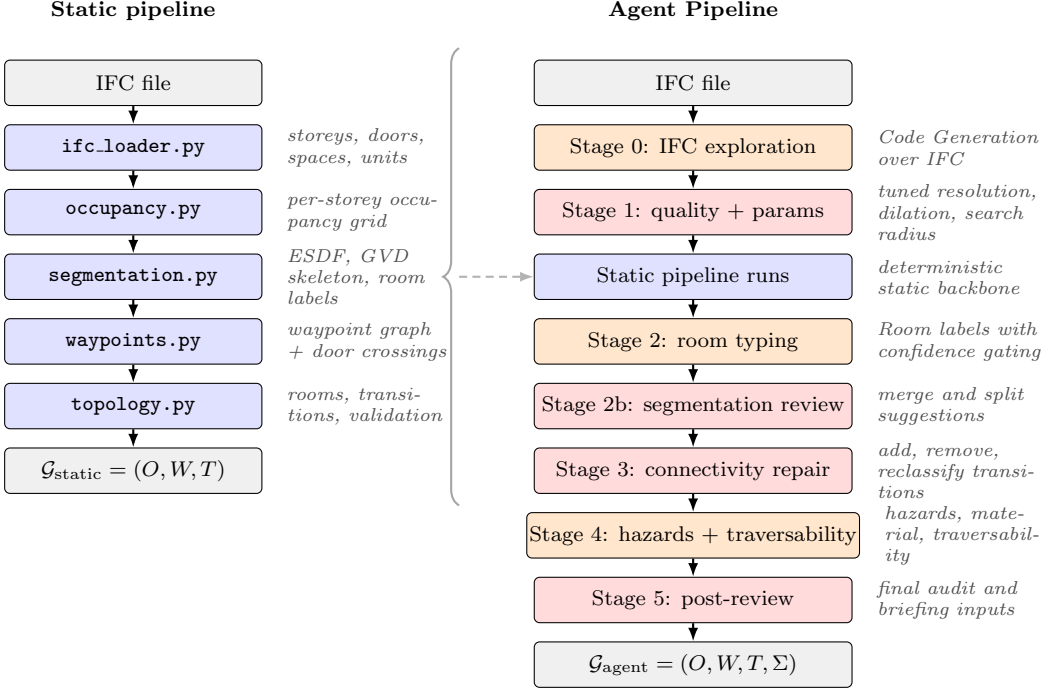

Static pipeline
IFC file
ifc_loader.py
storeys, doors, spaces, units
occupancy.py
per-storey occupancy grid
segmentation.py
ESDF, GVD skeleton, room labels
waypoints.py
waypoint graph + door crossings
topology.py
rooms, transitions, validation
$\mathcal{G}_{\text{static}} = (O, W, T)$
Agent Pipeline
IFC file
Stage 0: IFC exploration
Code Generation over IFC
Stage 1: quality + params
tuned resolution, dilation, search radius
Static pipeline runs
deterministic static backbone
Stage 2: room typing
Room labels with confidence gating
Stage 2b: segmentation review
merge and split suggestions
Stage 3: connectivity repair
add, remove, reclassify transitions
Stage 4: hazards + traversability
hazards, material, traversability
Stage 5: post-review
final audit and briefing inputs
$\mathcal{G}_{\text{agent}} = (O, W, T, \Sigma)$